

DOCUMENT #5 OF 10 — DATABASE ARCHITECTURE

Backend Schema & Data Integrity Model

Complete PostgreSQL Tables, Relationships, Indexing Strategies & Row-Level Security

Database Engine	Security Layer	Design Standard	Document Version
PostgreSQL 16 (Supabase Managed)	PostgreSQL RLS + JWT Claims	3NF Normalized + Fast B-Tree Indexes	v1.0 (Production Schema)

1. Authentication & Session Flow Lifecycle

- User Sign Up / Login:** Recruiter submits credentials or executes Google OAuth via Supabase Auth.
- Verify Credentials:** Supabase validates Argon2id hash or OAuth token.
- Create Session:** Record created in `auth.users` & session tracking row registered in `sessions` table.
- Access Application:** Client receives JWT containing `user_id` and `role` payload to query RLS-protected tables directly.

2. Complete Relational Database Tables

A. Core Entity Tables

Table Name	Schema Fields & Data Types	Foreign Keys & Constraints
1. users	id (UUID PK, gen_random_uuid()) name (VARCHAR(255)) email (VARCHAR(255) UNIQUE NOT NULL) password_hash (TEXT) role (VARCHAR(50) DEFAULT 'recruiter') created_at (TIMESTAMPTZ DEFAULT NOW()) updated_at (TIMESTAMPTZ DEFAULT NOW())	Syncs with Supabase auth.users. Role enum: super_admin, recruiter, hiring_manager.
2. projects	id (UUID PK, gen_random_uuid()) user_id (UUID NOT NULL) name (VARCHAR(255) NOT NULL) description (TEXT) status (VARCHAR(50) DEFAULT 'active') created_at (TIMESTAMPTZ DEFAULT NOW()) updated_at (TIMESTAMPTZ DEFAULT NOW())	FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE. Index on (user_id, status).
3. tasks (Job Screenings)	id (UUID PK, gen_random_uuid()) project_id (UUID NOT NULL) title (VARCHAR(255) NOT NULL) description (TEXT) status (VARCHAR(50) DEFAULT 'pending') due_date (TIMESTAMPTZ) created_at (TIMESTAMPTZ DEFAULT NOW())	FOREIGN KEY (project_id) REFERENCES projects(id) ON DELETE CASCADE. Index on (project_id).
4. candidates / resumes	id (UUID PK, gen_random_uuid()) task_id (UUID NOT NULL) name (VARCHAR(255)) email (VARCHAR(255)) phone (VARCHAR(50)) resume_url (TEXT NOT NULL) match_score (INT DEFAULT 0) ats_score (INT DEFAULT 0) status (VARCHAR(50) DEFAULT 'pending_review') created_at (TIMESTAMPTZ DEFAULT NOW())	FOREIGN KEY (task_id) REFERENCES tasks(id) ON DELETE CASCADE. Status: selected, shortlisted, pending_review, rejected.

B. Operations, Engagement & Billing Tables

Table Name	Schema Fields & Data Types	Foreign Keys & Constraints
5. comments	id (UUID PK, gen_random_uuid()) task_id (UUID NOT NULL) user_id (UUID NOT NULL) message (TEXT NOT NULL) created_at (TIMESTAMPTZ DEFAULT NOW())	FOREIGN KEY (task_id) REFERENCES tasks(id) ON DELETE CASCADE FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE.
6. notifications	id (UUID PK, gen_random_uuid()) user_id (UUID NOT NULL) title (VARCHAR(255) NOT NULL) message (TEXT NOT NULL) is_read (BOOLEAN DEFAULT FALSE) created_at (TIMESTAMPTZ DEFAULT NOW())	FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE. Index on (user_id, is_read).
7. sessions	id (UUID PK, gen_random_uuid()) user_id (UUID NOT NULL) token (TEXT NOT NULL) device_info (VARCHAR(255)) expired_at (TIMESTAMPTZ NOT NULL) created_at (TIMESTAMPTZ DEFAULT NOW())	FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE. Index on (token) for quick lookups.
8. payments	id (UUID PK, gen_random_uuid()) user_id (UUID NOT NULL) amount (NUMERIC(10,2) NOT NULL) currency (VARCHAR(10) DEFAULT 'USD') status (VARCHAR(50) NOT NULL) payment_method (VARCHAR(50)) created_at (TIMESTAMPTZ DEFAULT NOW())	FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE RESTRICT. Index on (user_id, created_at).

3. PostgreSQL DDL Implementation Script (with RLS Policies)

```
-- Enable UUID Extension
CREATE EXTENSION IF NOT EXISTS "uuid-osspl";

-- Projects Table & RLS
CREATE TABLE public.projects (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID NOT NULL REFERENCES auth.users(id) ON DELETE CASCADE,
    name VARCHAR(255) NOT NULL,
    description TEXT,
    status VARCHAR(50) DEFAULT 'active',
    created_at TIMESTAMPTZ DEFAULT NOW(),
    updated_at TIMESTAMPTZ DEFAULT NOW()
);
ALTER TABLE public.projects ENABLE ROW LEVEL SECURITY;
CREATE POLICY "Users view own projects" ON public.projects FOR ALL USING (auth.uid() = user_id);

-- Tasks (Job Screening Rounds) Table & RLS
CREATE TABLE public.tasks (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    project_id UUID NOT NULL REFERENCES public.projects(id) ON DELETE CASCADE,
    title VARCHAR(255) NOT NULL,
    description TEXT,
    status VARCHAR(50) DEFAULT 'pending',
    due_date TIMESTAMPTZ,
    created_at TIMESTAMPTZ DEFAULT NOW()
);
ALTER TABLE public.tasks ENABLE ROW LEVEL SECURITY;
CREATE POLICY "Users access tasks via project owner" ON public.tasks FOR ALL
USING (EXISTS (SELECT 1 FROM public.projects WHERE projects.id = tasks.project_id AND projects.user_id = auth.uid()));
```

4. Performance Tuning & Indexing Guidelines

- **Composite Indexes:** Add `CREATE INDEX idx_candidates_task_score ON candidates(task_id, match_score DESC)`; to ensure leaderboard ranking queries load in <10ms.
- **Data Integrity:** Strict Foreign Key constraints prevent orphan candidate comments or task files.
- **Automated Timestamp Updating:** Trigger attached to projects and users to auto-refresh `updated_at` on row changes.